

## Chapter 5

# HUMORTOOLS: AN ADAPTIVE WORKFLOW FOR A CREATIVE TASK

### 5.1 Introduction

Humor is a highly valued human skill. It is a sign of creativity and intelligence that enriches many domains: entertainment, advertising, social bonding, education and journalism (e.g. *The Daily Show*). Its engaging and surprising nature make it a compelling and memorable form of communication. As an intellectual challenge, humor has been studied by great Western thinkers – Plato, Kant, Freud [10] – and by modern philosophers, such as Daniel Dennett [71], and linguists [84, 83], all of whom have found humor difficult to describe.

Humor generation presents a grand challenge for artificial intelligence because it requires language abilities, world knowledge, and understanding of how people process information that computers do not yet have. Furthermore, many people find it challenging to create humor on demand. For those who can, it is a process that is almost entirely internalized and difficult to explain. This leads us to think that it requires the magical “black box” of creativity.

Generating humor in a systematic, externalized process requires decomposing it into simple steps. There is no obvious decomposition of humor so to approach the problem we take inspiration from philosophers’ and linguists’ descriptive models of humor. The descriptive models are often insightful. For example, the benign violation theory of humor postulates that humor violates our expectations [84]. This model feels intuitively true and insightful. Unfortunately, it is too abstract to apply directly. To generate humor, we need concrete mechanisms that embody abstract models. With the descriptive models in mind, we performed an analysis of 330 jokes from *The Onion* looking for concrete mechanisms that

fit or refined the descriptive models. Although these concrete mechanisms are simple, they still require people’s knowledge and language abilities to apply. Thus we propose a set of microtasks that can be used in a systematic and decomposed process for generating humor.

Traditional crowdsourcing uses microtasks in static workflows, such as *iterative improvement* or *find-fix-verify* [69, 19] to combine microtasks into a useful output. However, these simple patterns are insufficient for the difficult problem of humor creation. Instead, we must apply microtasks in an adaptive workflow that allows users more freedom to selectively choose microtasks that fit the current context of the problem.

This paper presents HumorTools, a systematic process for generating humor through microtasks combined in an adaptive workflow. HumorTools models the humor generation process as a constraint satisfaction problem that is met by searching a conceptual space. The constraint is to violate an expectation, and the conceptual space to search is defined by a 20 microtasks such as coming up with associations for words, writing down expectations in a given situation, and imagining the expectations and beliefs from unexpected points of view. The adaptive workflow asks workers to iteratively apply microtasks in a brainstorming-like fashion, until they can meet the constraint of violating an expectation. Not all chains of microtasks will lead to an expectation violation, so users may need to backtrack and follow a different chain of associations.

This paper makes the following contributions:

- a survey of humor research and humor writing processes described by professional comedians
- twenty novel microtasks, distilled from a large-scale analysis of professional news satire, that are useful for humor generation
- an adaptive workflow that poses humor generation as a constraint that can be satisfied by searching a conceptual space
- the HumorTools implementation of our adaptive workflow and microtasks, and an

evaluation of its performance on 20 people, finding that the 75% of users were able to generate humor in this externalized way, users reported that the workflow make their process methodical and enable them to make a wider variety of jokes. Three of their jokes were rated as funnier than *The Onion*.

## 5.2 Survey of Humor Literature

To inform our design of a microtask-based, humor-creation workflow, we wanted to derive vocabulary, theories, and process ideas from experts in humor.

### 5.2.1 Theories of Humor

Humor has been studied for thousands of years. It has been explored by Ancient Philosophers such as Plato and Aristotle, modern philosophers like Kant and Schopenhauer, and is an active field of study in linguistics and psychology. But by and large, these thinkers have attempted to define humor and to understand *why* things are funny, not to mechanize the humor-generation process. The three major categories of humor theories found in the literature are superiority, release, and incongruity.

- **Superiority theories** claim that humor is the result of feeling superior to somebody. Plato and Hobbes describe all humor as having a basis in insult, when the listener understands the insult, they can feel superior to the people being insulted.
- **Release theories** claim that humor is release of tension built up from the suppression of discussion in society. Freud is the the champion of these theories noting that taboo subjects are often the target of jokes particularly because of the emotional release they allow.
- **Incongruity theories** claim that humor comes from the realization of an informational anomaly. The idea dates back to Aristotle but has been the focus of most modern humor theories. Benign Violation Theory [84] claims that humor comes from detecting

an incongruity between expectations and reality, as long as the reality is not overly offensive. Kant, and later Schopenhauer, claimed humor was the result of perceiving a metaphor and thus seeing something in a new way that you would have previously thought unrelated.

Incongruity theories allow theorist to understand why jokes are funny by modeling the cognitive process of listeners. For example, according to Sul [84] (and others) an incongruity is not just an instance when expectations are violated, because there are many examples when expectations are violated that are not funny. To be funny, the incongruity must also *resolve* in the listeners mind. In Sul’s cognitive model of humor appreciation, a joke has a setup in which the reader predicts an expected outcome. At the end if the listener predicts the outcome, then there is no surprise and thus no laughter. If there is surprise, then there is incongruity. If the listener is confused, then the incongruity is not resolved and it is not funny. However, if the listener can find a logic that makes the (unexpected) ending follow from the set up, then it is funny.

The Semantic Script Theory of Humor (SSTH), presented by Raskin in 1985 [84], is a widely accepted incongruity model of humor. According to SSTH, each joke can be interpreted according to two distinct, opposing scripts. One of those scripts is usually an expectation based on the set up, and the other is often an inference based on the punchline. An important aspect of this theory is that the incongruity is often in the subtext of what is said, and not the text itself.

### 5.2.2 *Survey of Humor Advice from Experts*

While many people assume that humor is a completely subconscious process that cannot be taught, numerous comedians have written books claiming that humor is learnable and that there is a conscious process for creating it. We selected five popular books by comedians, describing aspects of their humor creation process. We refer to each book by the authors’ last name: Carter [26], Dean [33], Kaplan [50], Holloway [45], Vorhaus [96].

### *Structure: Setup and Punchline*

Jokes have set-up and punchline — the set-up establishes expectations and the punchline violates those expectations. The setup should be *relatable*, these are easy to describe in brief and will lead listeners to make many assumptions. Punchlines violate expectations by saying something that fits with the setup, but is unexpected. This aligns well with many incongruity theories of humor. Additionally, three of the books argue that the core of a punchline is truth. In summary, a good punchline is insightful — it helps you see something new about a familiar situation or object.

### *Exploration*

All the comedians agree that writing a joke requires exploration of a topic. Some explorations may lead to jokes, but many will fail, and you cannot know which ones will work in advance. This aligns well with the design literature on brainstorming and ideation. Holloway explicitly mentions mind maps, referring to them as “joke webs,” as a way to explore a topic. Exploration can be used at many stages of the joke writing process. Carter emphasizes its use to determine a good premise for a joke, while Dean discusses the need to consider many alternative punchlines for a given set up.

### *Details*

Jokes, like most good writing are richer when they contain details. Details make words more vivid to an audience, draw people more deeply into the work, and increase their emotional investment. Carter and Vorhaus both express a connection between details and truth. When a comedian strives for details, she has a better sense of the actual idea she is trying to portray. Carter advises that a premise for a joke should express a detailed picture of the emotion it is trying to portray.

### *Point of View*

All of the comedians reinforced the importance of point of view in constructing jokes. Most people naturally tell stories or jokes from their own perspective. But the material becomes more immediate when it is told from the point of view of the person directly affected. Here is an example of a joke that has had its point of view transformed from neutral to immediate:

Before:

*Q: How many Amish does it take to screw in a light bulb?*

*A: Two: one to screw in the light bulb, and one to wonder what it's for.*

After:

*Q: How many Amish does it take to screw in a light bulb?*

*A: What's a light bulb?*

In the second version, where the listener is assumed to be the Amish person, the punchline is shorter, more immediate, and funnier.

### *Assumptions and Inference*

Jokes are a story stripped down to its essentials. When writing a joke, the writer relies on the listener to make assumptions. Dean's example of this is the following: *"My wife just ran off with my best friend. Boy, do I miss him."* The assumption from the setup is that he misses his wife. The inference from the punchline is that he does not miss his wife, he misses his friend. Both key facts are unstated, they are assumed. Listeners will connect the dots that the writer omits. However, writers often need to write ideas in long form, before deciding the what to omit.

### *Takeaways*

The expert advice on humor writing confirms that humor can be at least partially if not fully externalized. Experts expressed both high-level techniques such as explicitly violating

an expectation as well as low-level techniques such as using details or point of view. This indicates that any generative model of humor will require both high-level structure and low-level techniques.

### 5.3 Humor Analysis

The advice from experts gives us a good vocabulary and guiding principles for developing a humor writing process. However, it lacks details. To fill in the details, we analyze professional humor to see what mechanisms they repeatedly apply in their jokes.

Humor takes on many forms. This analysis focuses only one type of humor that is well-suited to analysis and experimentation: a section of *The Onion* called *American Voices*. *The Onion* is a popular source of humor among liberal college students. *The Onion* is most famous for its fake news headlines satirizing American culture. However, they also have a section called *American Voices* which starts with real news headlines and satirizes them by writing fake “average American” responses. For example, Table 5.1 shows one of the real news headlines and three fake responses to it written by *The Onion*.

*American Voices* is well-suited to our analysis because the headlines can be seen as an input to the system and the jokes can be seen as the output. The job of the analysis is to find the operations that connect the headlines to the responses. Because *The Onion* has three jokes per headline we know there are several paths of operations between a headline and a joke.

#### 5.3.1 Finding Patterns in American Voices Patterns

We performed an analysis of 330 *American Voices* joke, in response to 110 real headlines. We found four general patterns:

##### *Pattern 1: Multiple References*

Jokes make multiple references to aspects in the headline. The references can be either direct or indirect. In the following joke, the joke makes four references to the headline.

Table 5.1: Example of the *American Voices* style of humor: a real news headline, a byline, and 3 jokes — fake “average American” responses.

|               |                                                                                                                                                                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Real Headline | Justin Bieber Baptized In NYC Bathtub                                                                                                                                                                                                                                                                           |
| Real Byline   | Pop star Justin Bieber was baptized in a friend’s bathtub this weekend after weeks of Bible study and church services, with celebrity blogs reporting that the 20-year-old sought spiritual guidance in an attempt to wash away his sins following a scandal in which videos emerged of him using racial slurs. |
| Joke 1        | “Oh my God! Can I lick the tub?”                                                                                                                                                                                                                                                                                |
| Joke 2        | “Great, now my teenage daughter’s going to be begging me for \$300 so she can reaffirm her devotion to God, too.”                                                                                                                                                                                               |
| Joke 3        | “Never let it be said that Bieber’s PR people aren’t bringing new ideas to the table.”                                                                                                                                                                                                                          |



Table 5.2: Example of the Multiple Connections pattern showing four concrete or oblique connections between aspects of the headline and things mentioned in the joke.

|                                        |                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| People Bending iPhones At Apple Stores | “I can’t believe people would just walk into an Apple store and start breaking things like it’s a Best Buy.” |
| People                                 | “I can’t believe <b>people</b> would...”                                                                     |
| Apple Stores                           | “...walk into an <b>Apple store</b> ...”                                                                     |
| bending iPhones                        | “...and start <b>breaking things</b> ...”                                                                    |
| Apple Store                            | “...like it’s a <b>Best Buy</b> .”                                                                           |

#### People Bending iPhones At Apple Stores

*“I can’t believe people would just walk into an Apple store and start breaking things like it’s a Best Buy.”*

Two of the references are direct — “People” and “Apple Store” appear in the headline and the joke. In contrast, the other two references are indirect: The headline mentions “bending iPhones,” and the joke references this as “breaking things.” The headline mentions “Apple Store” and the mentions “Best Buy.” Apple Stores and Best Buys are not the same, but they are clearly related. They are both electronics retailers — Apple Stores are at the high end and Best Buy is at the low end. Table 5.2 shows the four references in detail. Virtually all the jokes reference at least two aspects of the headline.

#### *Pattern 2: Association Types*

Indirect references in jokes fall into six prominent classes. We call these classes Association Types because they rely on the associative thinking skills of the joke writer. The “Apple Store” / “Best Buy” a type of association we call Alternative Thing because the two aspects are in the same domain, but are competitors or alternatives in that domain. Other examples

of Alternative Thing include:

**UConn** Holding ‘Football 101’ Clinic For Female Fans

*“Sure, I guess UConn’s course is fine if you couldn’t get into **Yale’s** football clinic.”*

**Baskin-Robbins** To Honor Veterans With ‘First Class Camouflage’ Ice Cream

*“This sure makes **Ben and Jerry’s** look like a bunch of flag-burning ISIS sympathizers.”*

Alternative Thing is one of six Association Types. Other Association Types are: Alternative Reaction (chicken prom corsage is expected to be gross/ a person would want one), Details (SkyMall / useless home products such as ‘Summer Savannah’ Backyard Garden Lion Pedestal), Insult (Breaking Bad / having a terrible finale), Personality Flaw (George Clooney / obsessed, delusional stalker fan). Detailed examples including the headline and joke are in Table 5.3.

### *Pattern 3: Belief Types*

Associations explain many elements in jokes but an association is only part of a joke. In addition to associations, jokes contain a belief that the speaker holds about an aspect or association. For example, the belief in the Ben and Jerry’s joke is that ‘First Class Camouflage’ Ice Cream” is bad for Ben and Jerry’s because it makes them look unpatriotic. This belief type is called Reason Bad. Another Joke with a Reason Bad belief is as follows:

McDonald’s Testing Customizable Burgers To Compete With Chipotle

*“Whatever happened to walking into a fast food restaurant, shouting a number, and eating whatever you were given?”*

Table 5.3: Examples of the Association Types pattern from our analysis of *American Voices* humor.

| Association Type                   | Headline                                                            | Headline Aspect      | Association                                      | Joke                                                                                                                      |
|------------------------------------|---------------------------------------------------------------------|----------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Alternative Thing                  | People Bending iPhones At <b>Apple Stores</b>                       | Apple Store          | Best Buy                                         | “I can’t believe people would just walk into an Apple store and start breaking things like it’s a <b>Best Buy</b> .”      |
| Alternative Person (Point of View) | <b>Justin Bieber</b> Baptized In NYC Bathtub                        | Bieber               | Bieber fan                                       | “Oh my God! Can <b>I</b> lick the tub?”                                                                                   |
| Alternative Reaction               | KFC Selling <b>Chicken Prom Corsages</b>                            | chicken prom corsage | person who would want this                       | “My daughter’s dress would look better with something from Burger King.”                                                  |
| Specific Example                   | Report: <b>‘SkyMall’ Magazine</b> May End Print Edition             | SkyMall              | ‘Summer Savannah’ Backyard Garden Lion Pedestals | “Alright, how many <b>‘Summer Savannah’ Backyard Garden Lion Pedestals</b> do I have to order to turn this thing around?” |
| Insult                             | Toys ‘R’ Us Pulls <b>‘Breaking Bad’</b> Action Figures From Shelves | Breaking Bad         | Finale was bad                                   | “That’s okay. My little guy <b>hated the finale</b> .”                                                                    |
| Personality Flaw                   | <b>George Clooney</b> Engaged                                       | George Clooney       | obsessed, delusional stalker fan                 | “ <b>He usually tells me everything</b> , so I doubt this is true.”                                                       |

The belief in this joke is that testing new burgers is bad. The reason it is bad is because people prefer things the traditional way, not the new way.

Reason Bad is one of eight belief types. Other belief types include Reason Good, Effect and Solution. Examples are in Table 5.4. “Effect” is the belief that an aspect of entity in the headline will have effect on the world or on the speaker of the joke. For example: .

Scientists Developing Heat-Resistant Chickens To Withstand Climate Change

*“How much to upgrade the chicken I already got?”*

The aspect ‘developing chicken technology’ could have an effect of requiring people to upgrade their technology.

The Solution belief finds a problem in the headline and proposes a solution to it. For example: .

Report: ‘SkyMall’ Magazine May End Print Edition

*“Alright, how many ‘Summer Savannah’ Backyard Garden Lion Pedestals do I have to order to turn this thing around?”*

The aspect of “Ending Print Edition” is a problem for this speaker and his solution is to buy more SkyMall stuff in order to “turn this thing around.”

#### *Pattern 4: Expectation Violation Types*

From the humor theory and practice we know that expectation violation play a role in humor generation and appreciation. Violating Expectations is an abstract goal, but our analysis of the *American Voices* found three concrete mechanisms for violating expectations.

#### **Violation Type 1: Bait-and-Switch**

In the Bait-and-Switch Violation Mechanism, the joke first grounds the listener in a statement that they expect to hear such as “I can’t believe people would just walk into an Apple Store

and break things”, but then abruptly adds a short phrase that changes the meaning entirely such as “like it’s a Best Buy.”

Here is another example of Bait-and-Switch:

Facebook Lets Users Choose Who Controls Their Account After Death

*“I’m not sure who I can trust with the solemn task of rejecting Bubble Witch Saga Requests.”*

The bait is ‘I’m not sure who I can trust with this solemn task’, and the switch is “rejecting Bubble Witch Saga Requests.”

The Bait-and-Switch mechanism is consistent with Semantic Script Theory of Humor in that the bait is the first script and the switch is the second script.

## **Violation Type 2: Unexpected Angle**

In the Unexpected Angle Violation Mechanism, the joke expresses an unexpected reaction to the headline by finding an unusual angle to interpret the headline through. Consider the following joke,

Great White Shark Populations Surging Off East Coast

*“It’s an exciting time to be a shark, that’s for sure.”*

The expected reaction to the headline is that the surging shark population is bad because it creates a danger. However, that interpretation is from the point of view of a human who is afraid of sharks. From the point of view of a shark this is a good thing. The shark’s point of view is the unexpected angle, and the unexpected belief is that this is a good thing. The belief that surging shark populations is a good thing violates expectations, which is why this joke is funny.

In the Camouflage Ice Cream joke, the usual angle is Ben and Jerry’s. Ben and Jerry’s is not directly mentioned in the headline, it is an Alternative Thing Association for Baskin-Robbins. The listener expects that releasing a new flavor of ice cream to honor veterans

is good, but it is bad when you consider the effect on the competitor: Ben and Jerry's. Considering the negative affect on an alternative aspect violates expectations.

### **Violation Type 3: Sarcasm**

In the Sarcasm Violation Mechanism, the joke a false assumption in the headline by agreeing with it to a degree that makes it clear to the reader that the speaker is not serious. For example,

Mick Jagger Blamed For Brazil's Historic World Cup Defeat

*"There is absolutely no other explanation."*

The headline contains the blatantly false assumption that Mick Jagger is to blame for Brazil's defeat. The joke agrees with this belief in such an exaggerated manner that it makes it clear the speaker is not serious, and thus brings the false assumption in the headline to the listeners attention. Sarcasm violates expectations in two ways: first, listeners do not expect people to say the opposite of what they mean and second, listeners are not fully aware of the false assumptions in the headlines thus highlighting them with a sarcastic response reveals an unexpected falsehood in the headline.

Here is another example of Sarcasm:

Baskin-Robbins To Honor Veterans With 'First Class Camouflage' Ice Cream

*"I look forward to placing a few of these on the graves of the fallen."*

This joke asserts that ice cream is a good way to honor veterans. However, this belief is sarcastic. Certainly in comparison with the traditional way of honoring veterans (placing flowers on graves), ice cream is an inappropriate way to honor veterans. The joke exposes the false assumption in the headline that Camouflage Ice Cream honors veterans.

Table 5.4: Examples of the Belief Types pattern from our analysis of *American Voices* humor.

| Belief Type | Headline                                                                         | Headline Aspect                     | Belief                                                                | Joke                                                                                                                       |
|-------------|----------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Reason Good | Baskin-Robbins To <b>Honor Veterans</b> With 'First Class Camouflage' Ice Cream  | Honor Veterans                      | This is good because ice cream is a great way to honor veterans       | " <b>I look forward to</b> placing a few of these on the graves of the fallen."                                            |
| Reason Bad  | McDonald's <b>Testing Customizable Burgers</b> To Compete With Chipotle          | Testing an improvement to fast food | This is bad because some people prefer things the way they are        | " <b>Whatever happened to</b> walking into a fast food restaurant, shouting a number, and eating whatever you were given?" |
| Effect      | Scientists <b>Developing Heat-Resistant Chickens</b> To Withstand Climate Change | developing chicken technology       | The effect on people who own chickens is upgrading their 'technology' | "How much to <b>upgrade</b> the chicken I already got?"                                                                    |
| Solution    | Report: 'SkyMall' Magazine May <b>End Print Edition</b>                          | End Print Edition                   | A solution to going out of business is to buy Skymall stuff.          | "Alright, how many 'Summer Savannah' Backyard Garden Lion Pedestals do I have to order to <b>turn this thing around</b> ?" |

## 5.4 System

### 5.4.1 Design

To simplify the problem of humor generation, we restrict ourselves to generating *American Voices* style of humor. The input to the system is a headline, and the output is a joke. As a further simplification, we use the same headlines that *The Onion* uses. There is a separate problem of selecting the headlines, that is left as future work.

We decompose the process of humor generation as a set of microtasks and a generative model of humor that uses those microtasks. The microtasks can be derived from the theory-inspired patterns found in the Humor Analysis section: breaking headlines into aspects, finding entity associations with aspects of the headline, defining new unexpected beliefs to express from the standpoint of associated entities. There are many possible ways to design a workflow that uses those microtasks. We first present two initial workflow designs that we

rejected before presenting the HumorTools workflow design.

#### 5.4.2 Initial Design: Static Workflow

Our first design was a static sequence of microtasks people could apply to a headline to generate a joke. We defined the static sequence by selecting a joke and then back-engineering a sequence of microtasks needed to generate it. Here is an example of a joke and a back-engineered workflow for it:

Report: Uber Adding ‘Panic Button’ For Chicago Passengers

*“It’s nice knowing Uber is willing to do everything it can to protect its customers short of properly screening its drivers.”*

Static Workflow:

1. Write down a expected reason this is good (“Uber is adding safety features for its passengers”)
2. Pick an aspect of the headline (“Uber”)
3. Write an insult about the entity (“Uber doesn’t properly screen its drivers”)
4. Combine the expected reason this is good from (1) with the insult from (3) into a Bait-and-Switch joke

When prototyping this workflow, we encountered two major problems. First, users were not always able to come up with the associations we wanted them to for new headlines or for the headline the pattern was derived from. An explanation for this is that users have limited world knowledge and it is unreasonable to expect them to come up with a specific kind of association for a given aspect. The system must allow enough flexibility for users to work with the limited world knowledge that they have.

Second, this static workflow did not generalize to many of the jokes we analyzed. Neither did any other static workflow we found. In fact, we found fewer than 10 jokes in the 330



that we analyzed that could be generated from the same static workflow. In order to write jokes that have the wealth of diversity that *The Onion* does, we have to be able to more flexibly apply the individual microtasks. The workflow cannot be static, it must adapt to the headline and to the users' knowledge.

#### 5.4.3 Initial Design: Brainstorming

Our second design was to use microtasks to brainstorm associations with the headline that would lead to a joke. Given a headline and a set of 6 Association microtasks, users were able to apply at least one association to any headline. However, after brainstorming 10-20 ideas users were unclear on how to write jokes. Although they had many ideas, none of them constituted a joke. Additionally, when we pushed them to write anything joke-like, they had trouble. They reported being overwhelmed by the number of ideas and had no guidance on how to create a joke out of them. This workflow had the problem that many brainstorming sessions have — that the ideas are good, but it lacks a path to a concrete outcome. Although brainstorming is useful technique, a generative model with more scaffolding is needed to complete the joke writing process.

#### 5.4.4 HumorTools Design

The descriptive theories of humor stress that expectation violation is the essence of a joke and this idea is born out in the three concrete mechanisms that *American Voices* uses in its jokes. One reason jokes are hard to write is because it is not trivial to violate someone's expectations using these mechanisms. Thus, in the HumorTools generative model, fulfilling one of the 3 expectation violation mechanisms is a constraint that must be satisfied. To help satisfy this constraint, the associations define a search space on concepts characteristic of *American Voices* humor. Multiple associations may be applied to satisfy the constraint, thus most microtasks can be chained together. Not all chains of microtasks are guaranteed to result in a joke. Thus HumorTools enables users to explore a tree of associations stemming from the headline, and they can backtrack when they get stuck.

#### 5.4.5 Implementation

HumorTools is a web application implemented in Meteor. It defines 20 microtasks and teaches them each in isolation by showing examples of professional headlines and joke with pieces missing, and asking the user to fill in the gaps. Users are not expected to get the same answers as *The Onion*, but they can see what *The Onion* wrote and self-assess whether they are thinking along the same lines.

Although microtasks are independent, they are presented in an order that emphasizes the patterns in which they are typically chained together. Identifying headline aspects is almost always first. This is typically followed by either writing an expected reaction to the headline, or by writing an association to an aspect of the headline. Belief microtasks typically follow from an Alternative Thing or Alternative Person Association. Unexpected Angle violation mechanisms can often be constructed from the belief of an Alternative Person Association.

When writing jokes for headlines, users apply microtasks at their own discretion. HumorTools support the writing process by structuring their exploration of the conceptual space in a tree. This facilitates backtracking, keeping multiple threads open, and knowing which aspects of the headline have not yet been explored. HumorTools instructs workers to record the constraints they met in their final joke. This gives them a heuristic for when they are done, and lets them know what they are missing and must keep working to complete.

#### 5.4.6 Microtasks

HumorTools teaching users 6 types of Microtasks, some with many subtypes of Microtasks. An example of the interfaces uses to teach the microtasks are given in Figure 5.1. In this section we detail the input and output of each microtask and provide examples.

##### *Identify Aspects*

In the Identify Aspects microtask, the user is given a headline, and the goal is to list individual aspects of it. For example, in the headline:

|                 |                                                                                                                                                                                                                                           |  |  |               |               |          |               |     |               |         |              |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------|---------------|----------|---------------|-----|---------------|---------|--------------|
| <b>Headline</b> | <b>Justin Bieber Baptized In NYC Bathtub</b>                                                                                                                                                                                              |  |  |               |               |          |               |     |               |         |              |
| <b>Aspects</b>  | <table border="1"> <tr> <td>Justin Bieber</td> <td><i>person</i></td> </tr> <tr> <td>baptized</td> <td><i>action</i></td> </tr> <tr> <td>NYC</td> <td><i>thing?</i></td> </tr> <tr> <td>bathtub</td> <td><i>thing</i></td> </tr> </table> |  |  | Justin Bieber | <i>person</i> | baptized | <i>action</i> | NYC | <i>thing?</i> | bathtub | <i>thing</i> |
| Justin Bieber   | <i>person</i>                                                                                                                                                                                                                             |  |  |               |               |          |               |     |               |         |              |
| baptized        | <i>action</i>                                                                                                                                                                                                                             |  |  |               |               |          |               |     |               |         |              |
| NYC             | <i>thing?</i>                                                                                                                                                                                                                             |  |  |               |               |          |               |     |               |         |              |
| bathtub         | <i>thing</i>                                                                                                                                                                                                                              |  |  |               |               |          |               |     |               |         |              |

|                                            |               |                          |                                                               |
|--------------------------------------------|---------------|--------------------------|---------------------------------------------------------------|
| <b>Headline</b>                            | <b>Aspect</b> | <b>Expected Reaction</b> | <b>Expected Reason / Belief</b>                               |
| Study: Too Much TV Can Lead To Early Death | early death   | bad                      | effect: we should stop watching so much TV if it's killing us |

|                                                                                                                                                                                                    |                                                                       |                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Headline</b>                                                                                                                                                                                    | <b>Joke</b>                                                           |                                                                               |
| Baskin-Robbins To Honor Veterans With 'First Class Camouflage' Ice Cream                                                                                                                           | I look forward to placing a few of these on the graves of the fallen. |                                                                               |
| <b>Expectations</b>                                                                                                                                                                                | <b>Violation</b>                                                      | <b>Sarcasm</b>                                                                |
| <ul style="list-style-type: none"> <li>This is weird</li> <li>Why? ice cream is an unusual way to honor veterans.</li> <li>Usually you honor veterans with flowers</li> <li>This is bad</li> </ul> | <b>Belief Type</b><br><b>Belief</b>                                   | This is good<br>Ice cream is a good way to honor veterans, just like flowers. |

Figure 5.1: Three exercises from the microtask tutorial. The highlighted text is typed by the user. Microtasks from top to bottom: Aspect, Expected Reaction and Reason, Violation.

*“Justin Bieber Baptized in NYC Bathtub”*

Aspects include: “Justin Bieber”, “Baptized”, “NYC”, and “Bathtub.”

This simple task is an easy way to start writing a joke for a headline, and helps expand the headline into pieces that can be reused in the joke. Some aspects will be used directly, and some will undergo transformations from other microtasks.

*Expected Reactions and Reasons*

In the Expected Reaction and Reason microtask, the user chooses an aspect, and writes down whether they expect the reaction to be positive or negative and why. For example, one reaction to “Justin Bieber” is that he is bad. Possible reasons include disliking his music or that he is a bad role model.

Jokes are characterized by violating expectations. Thus, writing down expected reactions makes it clear what the expectations the user is going to violate are. Giving a reason helps expand an intuitive reaction into more material to find opportunities to violate the expectation.

*Aspect Associations*

In an Association microtask, the user chooses an aspect and writes down an association with it. For example, “Justin Bieber” is associated with “obsessive fans.” People’s associative memory makes brainstorming associations fairly easy. However, there are many possible ways to associate things, and some of them may be more useful to humor generation than others. In the analysis of *The Onion*, we identified six types of associations that *The Onion* uses repeatedly in their jokes. The “Justin Bieber” / “Justin Bieber fan” association is an Alternative Person type of association. Other Alternative Person associations are “Justin Bieber public relations people” or “Justin Bieber fan’s parent.”

Associations help expand the context of the headline to find more opportunities to violate expectations. Alternative Things, Alternative Person, Alternative Reaction are especially

helpful for finding point of view, or an angle to fulfill an Unexpected Angle expectation violation.

### *Beliefs*

In a Belief microtask, the user takes an aspect or associated aspect, and writes down a belief about it that is different than the expected reaction or reason. Here are examples of how three different Alternative Person associations can lead to three different Beliefs: A Justin Bieber fan would believe his baptism is a good thing, which is counter to the expectation that this it is bad. A Bieber fan’s parent would think it is a bad thing, but not for the expected reason — but because of the negative effect on them (“Great, now my teenage daughter’s going to be begging me for \$300’ so she can reaffirm her devotion to God, too.’). Lastly, from the standpoint of Bieber’s public relations team, this is good because it shows they are clever.

Beliefs are needed to take an association and make a story around it that will violate an expectation. This is consistent with the Semantic Script Theory of Humor that claims jokes need to have an expected script and an opposing script. Beliefs help users generate the opposing script.

Beliefs are fairly abstract, and like associations, there are many possible beliefs. From our analysis of *The Onion* we found eight Belief Types that were repeatedly used in jokes. The two most intuitive Belief Types are Reason Good and Reason Bad. Effect on the Speaker and Solution are two less intuitive but useful. See section 5.3.1 and Table 5.4 for examples. Other Belief Types include Effect on Others, Blame, Predict Fail, Sympathize.

### *Expectation Violation*

In an Expectation Violation microtask, the user connects pieces of information that complete one of the three Expectation Violation Mechanisms in section 5.3.1 and identifies the type mechanism which they are using and fills in specific details about that type of mechanism.

Because of the complexity of this microtask, we only taught users two of the three violation mechanisms we found: Unexpected Angle and Sarcasm.

In Unexpected Angle, users identified the angle (e.g., SkyMall shopper), Belief Type (e.g., Solution) and the Belief (e. g. That buying more SkyMall stuff will solve the problem).

In Sarcasm, users had to identify the hidden false assumption, the Belief Type, and the Belief. Consider the following joke:

Poll: Elite Colleges Don't Produce Happier Graduates

*"Is there no joy in this world for the privileged?"*

The hidden false assumption is that news is bad. The Belief Type is Sympathize. The Joke goes along with the false assumption that this is bad news by sympathizing with the elite. The Belief that expresses that sympathy is the exaggeration that there is no joy for the elite.

### *Heuristic Evaluation*

It is difficult to know when a joke is complete. We declare a joke complete when it passes the heuristic evaluation containing three things:

1. An Expectation Violation Mechanism (e.g. the Unexpected Angle Expectation Violation Mechanism)
2. A belief (e.g. the belief that Bieber's baptism is a good thing)
3. Two references to the headline (e.g. Bieber/Bieber fan is the first reference, and Bathtub/tub is the second reference)

Once the user writes their joke, they fill out a small form detailing a heuristic evaluation of the joke. For example, in the Bieber bathtub joke, the Expectation Violation mechanism is Unexpected Angle, the Belief is that Bieber's baptism is a good thing, and the two references are "Bieber" / Bieber fan (represented as the speaker of the joke, and "Bathtub" / "tub".

If any of these items are not in the joke, the user must backtrack and continue working. This heuristic evaluation is what ensures that the constraint has been met. Previously, we described that the jokes were about satisfying a constraint. That constraint actually has two parts: one is violating an expectation, and the other is containing two headline references. Satisfying the Expectation Violation constraint is much harder, so for simplicity we say that is the major constraint. Usually the second constraint is met by simply applying both an Association and a Belief. The Expectation Violation Constraint is inspired by the humor literature, the Two Reference Constraint comes from the analysis of *The Onion*, but some comedians mentioned that connections are a part of what makes something funny.

#### 5.4.7 Adaptive Workflow

Once users have been trained in the microtasks, they can start with a headline, explore a conceptual search and find an Expectation Violation. Users have must be allowed freedom in their choice of microtasks in order to adapt to the current context and to the limited to their knowledge. However, the search space is large and there are three techniques we use in the adaptive workflow to help guide the search and manage the exploration.

Although the microtasks are independent, there are some implicit constraints on their order. The first microtask is always to identify the aspects of the headline. This gives users a concrete way to start the joke writing process and sketches out the different branches to explore. Once aspects have been identified, it is typical to apply Association and Expectation microtasks to them. Additionally, there are pairs of microtasks where one microtask naturally transitions into the other. For example, Alternative Person Associations often leads to an Unexpected Angle Expectation Violation Mechanism. These transitions are not explicitly taught to users, but implicitly implied in the order of the microtasks we teach and how examples build on one another.

As a measure against users being overwhelmed by the size of the search space, HumorTools instruct users to follow their train of thought and if they get stuck, backtrack. People are much better at following one train of thought than jump between different contexts. However,

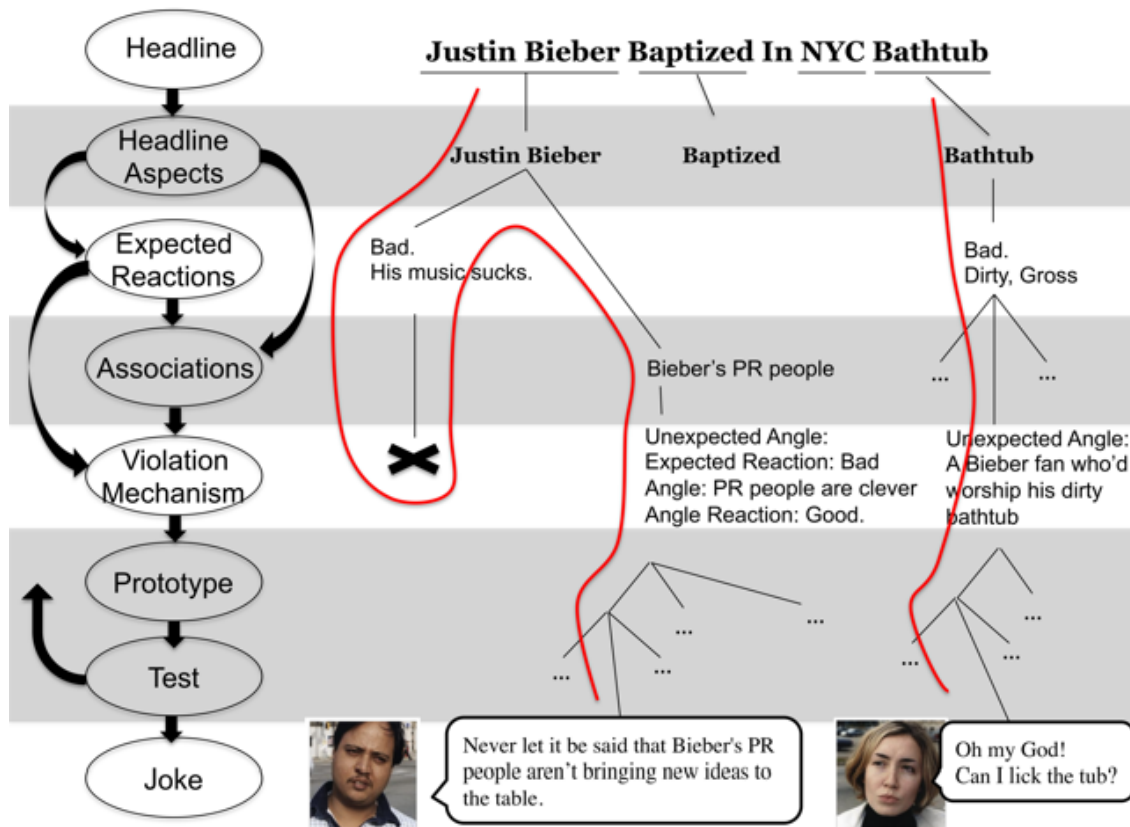


Figure 5.2: An illustration of the HumorTools adaptive workflow generating two jokes — one with backtracking, one without. The left side shows the typical transitions between types of microtasks, the right side shows the search paths through the conceptual search space from the headline input to the joke output.



people also have tendency to follow one train of thought too far. Although this is still a danger in any system with people in the loop, HumorTools’ design of writing out a branch explicitly makes it more attractive to backtrack and take a difference approach when the user gets stuck.

Users of HumorTools use the Heuristic Evaluation microtask to know when their joke is complete. This provides a stopping condition for the workflow. The Heuristic Evaluation looks for three things: an Expectation Violation Mechanism, a Belief, and Two References. It is up to the user to identify these aspects of their joke. If they can, they they are done, if not, they have to backtrack and either add to their joke or start a new one.

An illustration of the HumorTools search process is depicted in Figure 5.2. On the left are the key types of microtasks, arrows between the microtasks represent typical transitions. The right side shows the pathways for two jokes — one with backtracking and one without. Red lines indicate the two paths. The process starts by identifying aspects in jokes. For the first joke, starting with the “Justin Bieber” aspect we write the Expected Reaction (that he is bad) and a reason (“his music sucks.”) In this thread we did not see an opportunity to apply a violation mechanism, so we backtrack. Going back to Justin Bieber, we see opportunity to apply an Alternative Person Association. An overlooked person in this scenario are Bieber’s public relations team. From this Alternative Person Association there is a possibility to apply an Unexpected Angle Association. We expect this headline to be bad, but from the point of view of Bieber’s PR team, this is a good thing because they are doing their job. A belief we can apply to to Praise Bieber’s PR people for being clever. The joke here is “Never let it be said that Bieber’s PR people aren’t bringing new ideas to the table.”

The HumorTools Adaptive Workflow is not a deterministic process. Multiple jokes are possible. Another joke stems from the Expected Reaction to the Bathtub being dirty and gross, but the Unexpected Angle is that a fan would not care and would worship him anyway. The joke is “Oh my God! Can I lick the tub?” (Note: these jokes were written by *The Onion* to illustrate in an abbreviated fashion how these jokes could have been made. The Evaluation section has examples of jokes made by users.)

## 5.5 Evaluation

### 5.5.1 Setup

To evaluate HumorTools, tested if humor novices could use the microtasks and workflow to create *American Voices* jokes. From our early investigations, we knew that people can write *American Voices* jokes, but do so by intuition and with a limited range of techniques. Often they get stuck after one joke. Our aim was to teach people to write jokes using an externalized process. If that is possible, we want to know which elements of the workflow are most helpful to users, and how funny those jokes are compared to *The Onion*.

### 5.5.2 Study

We designed a web-based study lasting between 60-80 minutes that participants completed at home. We advertised the study through mailing lists on a large college campus. Participants were paid \$20 for their participation. The stages of the study were as follows:

1. Rate *American Voices* jokes. This gives users a warm-up and 24 examples of the target style of humor. This lasts about 5 minutes.
2. Write jokes intrinsically. We give participants 5 minutes to write jokes for 3 headlines. This gives users a chance to experience joke writing without the tutorial so they will later be able to compare their experiences. (referred to as the Intrinsic stage)
3. Do the tutorial. The tutorial on the microtasks and workflow. It is self-paced and lasts 30-50 minutes.
4. Write jokes with the HumorTools Workflow. We give participants 15 minutes to write jokes for the same three headlines as in stage 2. This gives users a chance to write compare their intrinsic joke writing process to the HumorTools workflow.

5. Survey. An exit survey asking about their experience and demographics. This lasts about 5 minutes.

In our study design, the Intrinsic joke writing step was only 5 minutes whereas the HumorTools writing steps is 15 minutes. These parameters were established in earlier prototypes. When writing joke intuitively, people do not need much time. However, in the HumorTools workflow, writing steps out takes time, so we allot more time in HumorTools to create a fair comparison.

In the Intrinsic and HumorTools joke writing stages, we used the same headlines as prompts. This means that when they see the headlines in the HumorTools writing session, they are already familiar with the headlines and have already written a joke intuitively for them. In some sense they are biased by already this familiarity, however that is not a problem for this study. First, it is good that they have already written a joke intuitively for the headline. In a non-experimental setting for joke writing using HumorTools, we do not expect to preclude people from writing jokes intrinsically. We expect that users would write jokes the intrinsic way until they ran out of ideas, then switched over to HumorTools. Thus, having already written intuitive jokes is consistent with intended practice. Second, using the same headlines in the Intrinsic and the HumorTools writing sessions is the only way to compare the two methodology. Because of the dynamic nature of joke writing, the choice of the headline determines much of the experience. People find some headlines harder or easier for reasons yet unknown to us. We want to make sure the difference in experience is due to the tools and not the headline choice. Randomizing for headlines is part of our future work.

The jokes were rated by the authors as either below the bar or above the bar for joke style and quality consistent with *The Onion*. Nine of the good jokes were selected by the authors to be rated for funniness against the *The Onion* by an audience of readers who enjoy *The Onion*. In this study, raters see a headline and six jokes for it. Three are written by *The Onion*, and 3 are written by HumorTools users. For each joke, the raters rate each joke by answering whether it is funny: “yes,” “no,” or “meh.”

### 5.5.3 Results

The humor writing study had 20 participants (8 female, 6 male, 6 declined to say) all between the ages of 18 and 25. The humor rating study had 21 participants, all of whom were college students who were fans of *The Onion*.

#### *HumorTools Completion*

The HumorTools process is novel and has many new concepts for users to learn in the tutorial. The biggest danger in HumorTools is users not completing the process. Across the twenty users in the study, we found that 75% of users were able to write complete jokes in HumorTools. 25% were above our quality bar. Of the 9 funniest jokes written in HumorTools, 3 were rated funnier than a professionally written Onion joke. See Figure 5.3. Although we do not know how much time *The Onion* put into their jokes or how many they rejected before deciding to print these, this represents a promising future for novice externalized humor generation processes.

#### *User Experience*

We asked users to compare their intrinsic humor writing approach with HumorTools. We wanted to know which they preferred and what was most helpful to them.

Of the 20 participants, 17 (85%) said that HumorTools was helpful to their joke writing process. Three (15%) said that the workflow was not helpful and they preferred to write jokes intrinsically.

For the participants who liked HumorTools, 6 of the 17 described their workflow as making joke writing more structured, systematic, or methodical. 5 of the 17 described their process before as being “*instinctual*” or they “*didn’t think about it*” or in one case their process before was to “*[let] things sit a lot more in my head and waiting for a joke to magically pop out*” (p5). Three participants said the workflow was helpful in getting started writing jokes.

The participants who reported disliking the workflow all had different reasons. One

## PETA Seeks Copyright for Primate

A lawsuit filed by PETA claims that “monkey selfies” snapped by a macaque who stole a photographer’s camera should be considered the legal property of the macaque himself.



*This is why you always get the macaque to sign a release.*



47% Funny



*This is great news for animal rights. Now my neighbor can take my dog to court instead of me the next time he deposits some of his “legal property” in their front lawn.*

62% Funny

## Liquid Water Found on Mars

NASA revealed Monday that they have found evidence of liquid water on the Mars, pointing to the possibility of life on the red planet.



*I think it's more convenient for me to get it from the tap.*

45% Funny



*In other news, Pluto has been called even less of a planet now.*

45% Funny



*Sounds refreshing!*



15% Funny

Figure 5.3: Two headlines with jokes from both HumorTools and *The Onion*. These are 3 instances where HumorTools jokes were rated funnier than *The Onion*. The two jokes written by *The Onion* have *Onion* logos on their far right.

thought it was better to just learn by example and another said it was over-simplified and “*only for beginners.*” In earlier pilots, we ran HumorTools on two users who had written professional or semi-professional humor. Both reported that HumorTools did nothing for them. One said it was easy enough to do intrinsically, the other said, “This is basically similar to the approach I use in my head, but [HumorTools] slows me down.” These are valid reasons not to use a workflow.

The two microtasks participants reported most helpful were Associations (10 of 17) and Violations (10 of 17). Aspects were mentioned by 4 participants, Expected Reactions were mentioned by 4 participants, Beliefs by 2 participants (all out of 17 participants who found the microtasks useful). This wide spread is probably product of users’ variety in innate humor writing abilities.

Associations were useful because they gave participants more possibilities for ideation. In their words, “[Associations were] helpful for finding a new angle or a way to be sarcastic” (p14) and “[Associations] helped me think of jokes in a wider conceptual space than I previously had.” (p9) With and without the workflow this microtask was helpful: “[Associations] generate a bunch of great ideas for a joke and makes it much easier to finish them with either the other techniques or just by thinking.” (p4)

Violations were useful because it make the actual mechanism for the joke more concrete and externalized. “I used the Violations because I knew that I had two choices — helped to narrow it down and make it feel manageable” (p14). Particularly, of the two Violation Types — Angle and Sarcasm — Angle was mentioned by 4 times participants who benefited from Violations. Their natural inclination was toward sarcasm and adding angle as a second mechanism enabled them to land far more jokes. Even for people who already had natural sarcastic tendencies, other microtasks enhances their sarcastic abilities when put together in this workflow. “Expected Reactions and Expected Beliefs [were useful] because they helped me to be sarcastic more intentionally — I could use them to make sure the jokes violated expectations” (p14).

All the microtasks were named as useful by at least one participant. It is not surprising

that participants found different things useful. We attribute this to their difference in inherent abilities to write humor. It is hard to know which microtasks will unlock possibilities for people and which ones they either do not need or already know. In the future, being able to rapidly assess this would make teaching faster and more effective.

### *Areas for Improvement*

We asked participants for areas of improvement. Four people found the tutorial too long and this probably impaired their ability to write jokes at the end due to exhaustion. Six people found one or more of the microtasks in the tutorial obvious. It is future work to teach the HumorTools microtasks more iteratively and more interactively.

## **5.6 Discussion**

HumorTools is a first step towards a system that makes humor generation externalized. HumorTools decomposes humor into microtasks and externalizes the process, but does not yet crowdsource the generation of humor. In the current study, one person writes a joke from start to finish. We believe that by decomposing humor generation into microtasks, we open the possibility of collaboratively writing jokes where multiple users can search the conceptual space of jokes together and build on the associations of others.

There are several ways to continuously improve the quality of the HumorTools jokes. The current set of microtasks is sufficient for writing humor, but we may be missing critical microtasks that *The Onion* uses to be funny. In fact, this is definitely true because we only taught three Expectation Violation Mechanisms found in *The Onion*. We can continuously improve the HumorTools workflow by comparing HumorTools jokes against *The Onion*, finding cases where *The Onion* is better, and analyzing those jokes for microtasks not yet in the tool.

HumorTools currently focuses on one type of humor, and it is future work to generalize the techniques to other forms of humor. Another form of humor that has input/output pairs is *The New Yorker* Caption Contest. *The New Yorker* has a different style of humor than

*The Onion*. To extend HumorTools to this new domain, we would need to analyze examples from *The New Yorker* to find what types of associations, beliefs, and expectation violation mechanisms they use. We can also possibly extend to forms of humor that do not have input/output pairs. Most of *The Onion* is satirical headlines — for which we have no input. However, we hypothesize that every joke has some kind of input or inspiration. Often it is hidden or diffuse, but by looking at many examples of *Onion* headlines, we could possibly back-engineer the types of targets that inspire their humor.

We believe that posing creative tasks as a constraint-satisfaction problem within a conceptual search space can be generalized to other domains. One such domain is writing persuasive argumentation, such as op-eds in newspapers. Extending this approach to other domains requires defining the constraints and search spaces that characterize the domain. This can be found by look at examples and drawing from theory. For persuasive argumentation, existing op-eds can serve as the examples and the study of rhetoric can provide clues as to what the constraint and search spaces might be. The most important constraint to meet in argumentation is to have a thesis that can be supported by arguments. This is generally how argumentation is taught. To find out what is persuasive, we could look at metaphors, tone, and the use of story narrative to make data fit into a compelling story.

## 5.7 Summary

Many people assume that humor requires the magic of creativity, and that it cannot be broken down and made systematic. HumorTools presents an externalized process for generating humor that relies on human intelligence, but makes it easier to write humor. Humor generation, and creativity in general, probably cannot be reduced to a static, cookie-cutter approach. The generative model we propose is a constraint problem that can be satisfied by searching a conceptual space of aspects, associations and beliefs. The HumorTools model was informed by theories of humor which characterize humor as violation of expectations, and by analysis of examples to find concrete mechanisms for violating expectations as well as microtasks that define the conceptual search space.



In a 20-person study of HumorTools 75% of users were able to generate humor in this externalized way. In a survey and interviews, users reported liked that it helped them start writing jokes in a concrete way, and gave the enough guidance to make the process more concrete. It also helped them create a wider variety of jokes than their internalized process did. In a comparison against *The Onion*'s jokes for the same headlines, three of the HumorTools jokes were rated funnier than *The Onion*. We believe this approach to decomposing creative problems into an externalized approach can generalize to other domains, lead to better ways to teach creativity, enable better collaboration, and result in more systematic generation of creative artifacts.